# P.E.N.Y

PERSONAL EXPENSE & NET-WORTH YIELD

# Build Guide

WINDOWS EDITION

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Part of the ALIS OS suite

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This is the version for your dad — a normal Windows program, no server, no Docker, no network setup. It's the exact same app and database format as the TrueNAS server edition, just packaged to run standalone.

## Important — I can't build the .exe myself, but here are two ways to get one

I've written and tested all the code, but I'm running in a Linux sandbox with no Windows machine and no reliable way to cross-compile a Windows installer from here — I checked, and the tools for that (like Wine) aren't a safe bet for something as fiddly as a native database module. It needs to be built somewhere that actually runs Windows. Two options:

### Option A — GitHub Actions (no Windows PC needed at all)

This folder includes `.github/workflows/build-windows.yml`, which builds the installer automatically on a real Windows machine that GitHub provides for free.

1. Create a GitHub account if you don't have one, and create a new repo (private is fine)
2. Upload this whole `peny-windows` folder to it
3. Go to the **Actions** tab on your repo — it starts building automatically
4. When it finishes (a few minutes), click the run, then download **peny-windows-installer** from the bottom of the page — that's your `.exe`

Easiest option if you don't have a spare Windows PC to build on.

### Option B — Build it yourself on a Windows PC

On a Windows PC with Node.js installed (get the LTS version):

1. Open this folder in a terminal (PowerShell or Command Prompt)
2. Install everything: `npm install` This also rebuilds the database engine for Windows automatically.
3. Build the installer: `npm run dist`
4. Look in the new `dist` folder for `P.E.N.Y Setup 1.7.3.exe` — that's the file to give your dad, complete with the P.E.N.Y icon, desktop shortcut, and taskbar entry.

## Security — what to expect, and what I've done about it

**When your dad runs the installer, Windows will probably show a blue "Windows protected your PC" SmartScreen warning.** This isn't a sign anything's wrong — it just means the installer isn't digitally signed by a registered publisher (that requires an annual code-signing certificate, usually £70–£300/year, which isn't worth it for a personal family app). To get past it: click **"More info"**, then **"Run anyway"**. This is completely normal for small, independently-built Windows apps.

**Some antivirus software may also flag it or scan it slowly the first time.** Same reason — it's new and unsigned, not because anything's actually wrong. If you want extra peace of mind before handing it over, you can upload the `.exe` to virustotal.com yourself and check it against 70+ antivirus engines first.

**On the app itself, two things worth knowing:**

- The app only listens on `127.0.0.1` (his own PC), not his whole home network — nobody else on his WiFi can reach it, unlike the TrueNAS server edition which is deliberately reachable across your network.

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- Login sessions are signed with a secret that's generated once and saved in his data folder, so he won't get logged out every time he reopens the app — but it does mean that folder should be treated like any other sensitive personal file (not shared, not synced to a public cloud folder unencrypted).

Beyond that, this is the same login (bcrypt password hashing), optional 2FA, and recovery-code system as the server edition — nothing about running it standalone on his PC makes the app itself less secure than your TrueNAS setup.

### Optional — getting rid of the SmartScreen warning entirely

**This is entirely optional and not something I'd recommend doing right now** — it's here for reference if the warning ever bothers you enough to act on it.

There's no genuinely free certificate that would work for a private app like this today. Free code-signing programs (SignPath Foundation, OSSign) only cover public open-source projects with an existing track record — a public repo, an OSI-approved licence, and (for OSSign specifically) at least 6 months of active project history. If P.E.N.Y ever gets its planned open-source GitHub release and sits public for a while, it's worth revisiting then.

The cheapest realistic paid route today is **Microsoft Azure Trusted Signing** (recently renamed Artifact Signing) — about $9.99/month, pay-as-you-go, so you could sign the installer for one month and cancel. It requires uploading ID for identity verification. Worth knowing: **even a paid certificate doesn't remove the warning instantly** — Microsoft discontinued the "instant trust" behaviour for EV certificates in 2024, so SmartScreen reputation now has to build up through real downloads over time regardless of certificate type.

Given this only affects one person (your dad, once), clicking "More info → Run anyway" is genuinely the simplest and most proportionate answer — but the option's documented here if that ever changes.

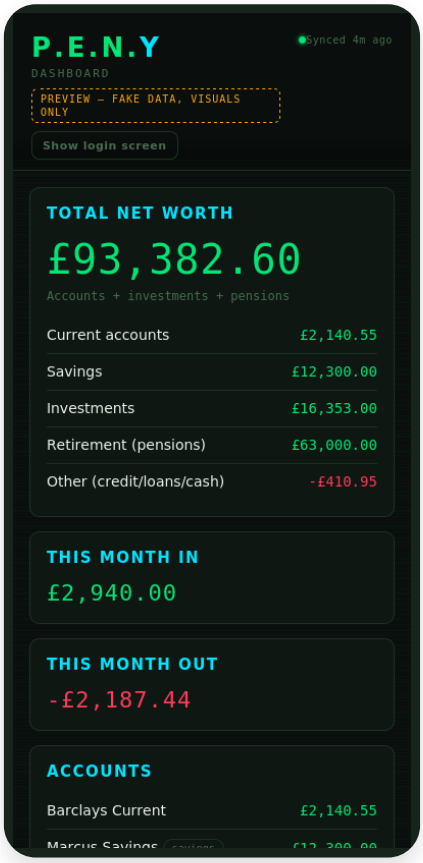
## What's in this folder

- `main.js` — starts the app when it's launched
- `backend/` — the same P.E.N.Y server code used on TrueNAS
- `frontend/` — the same app screens
- `icon.ico` — the P.E.N.Y app icon, already wired into the build
- `.github/workflows/build-windows.yml` — the automatic build (Option A above)
- `package.json` — tells the Windows installer builder what to do

## What happens when your dad runs it

- He runs the installer once, like any other Windows program
- P.E.N.Y appears in his Start Menu and on the Desktop, with its own icon
- Opening it starts the app in its own window — no browser needed
- If he ever forgets his password, there's a recovery code shown once at sign-up — see the note on password resets below
- All his data is saved automatically in his Windows user folder ( `C:\Users\<his name>\AppData\Roaming\P.E.N.Y\data` ) — back this up the same way you'd back up any other important folder (OneDrive, external drive, whatever he already uses)

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The Dashboard — the same app he'll see, running standalone on his own PC

## Password resets (no email server needed)

P.E.N.Y doesn't rely on an email server to reset a forgotten password — that would mean setting up SMTP on his PC too, which defeats the point of keeping this simple for him.

Instead, when he first creates his account, P.E.N.Y shows a one-time **recovery code**. He should write it down or print it and keep it somewhere safe (like a physical notebook, not saved as a file on the same PC). If he ever forgets his password, the "Forgot password?" link on the login screen asks for his email, that recovery code, and a new password — no internet connection or email account required. If he loses both his password and the recovery code, the only way back in is for you to open his `peny.db` file directly and reset it for him — worth keeping a copy of the recovery code yourself too, just in case.

## Updating it later

Since this shares the same code and database format as the TrueNAS version, when a new version comes out:

1. Rebuild a new installer the same way (`npm run dist`)
2. Send your dad the new installer — running it over the top keeps his data
3. Or, if you'd rather move his data to your TrueNAS server instead one day: **More > Export All Data (JSON)** on his machine, then **More > Restore a full P.E.N.Y backup** on the server version — everything carries across exactly.

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